Evaluation of increment PDFs using Laplace's method

1 Starting point: the distribution of h

We start from Equation (11) of [1], which defines the probability density function of the local Hölder exponent h:

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)}}{\displaystyle\int_{h_{\min}}^{h_{\max}} \left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)} \, \mathrm{d}h}. \tag{1}$$

Using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h-c_1)^2}{2c_2},$$

with $h_{\min} = -\infty$ and $h_{\max} = +\infty$, we obtain

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{(h-c_1)^2}{2c_2} \ln\left(\frac{L}{\ell}\right)\right]}{\displaystyle\int_{-\infty}^{+\infty} \exp\left[-\frac{(h-c_1)^2}{2c_2} \ln\left(\frac{L}{\ell}\right)\right] \mathrm{d}h}. \tag{2}$$

To simplify notation, we define

$$\alpha = -\ln\left(\frac{\ell}{L}\right) > 0. \tag{3}$$

Then the expression becomes

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right]}{\displaystyle\int_{-\infty}^{+\infty} \exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right] \mathrm{d}h}. \tag{4}$$

The denominator is a Gaussian integral:

$$\int_{-\infty}^{+\infty} e^{-\frac{\alpha}{2c_2}(h-c_1)^2} \mathrm{d}h = \sqrt{\frac{2\pi c_2}{\alpha}}. \tag{5}$$

Hence,

$$\boxed{\mathcal{P}_h^{(\ell)}(h) = \sqrt{\frac{\alpha}{2\pi c_2}} \exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right].} \tag{6}$$

2 Computation of the PDF using Laplace's method

2.1 General form of the PDF

We now consider the probability density function of velocity increments:

$$\mathcal{P}_{\delta_\ell u}(\delta_\ell u) = \frac{1}{2\pi\sigma} \sqrt{\frac{\alpha}{c_2}} \int_{-\infty}^{+\infty} \exp[-f(h, \alpha, \delta_\ell u)] \, \mathrm{d}h. \tag{7}$$

with

$$f(h, \alpha, \delta_\ell u) = \alpha\left(\frac{(h-c_1)^2}{2c_2} - h\right) + \frac{1}{2}\left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha h} \tag{8}$$

Laplace approximation Let $I = [a, b]$ be an interval, $\varphi \in C^2(I, \mathbb{R})$ and $g \in C^0(I, \mathbb{C})$. Assume that:

(i) For all $t > 0$, $\displaystyle\int_a^b e^{t\varphi(x)}|g(x)|\,dx < +\infty$;

(ii) φ' vanishes at a unique point $x_0 \in I$ and $\varphi''(x_0) < 0$;

(iii) $g(x_0) \neq 0$.

Then

$$\int_a^b e^{\,t\varphi(x)}\,g(x)\,dx \sim \sqrt{\frac{2\pi}{t\,|\varphi''(x_0)|}}\,e^{\,t\varphi(x_0)}\,g(x_0) \qquad \text{as } t \to +\infty.$$

Knowing that

$$\forall h \in \mathbf{R}, \quad \frac{\partial^2 f}{\partial h^2}(h, \alpha, \delta_\ell u) = \frac{\alpha}{c_2} + 2\alpha^2 \left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha h} \quad > 0 \tag{11}$$

we just need to verify that $h^\star$ is such that $\frac{\partial f}{\partial h}(h^\star, \alpha, \delta_\ell u) = 0$ to be able to use Laplace approximation.

In this case, by applying the approximation to Eq (7), we get :

$$\boxed{\mathcal{P}_{\delta_\ell u}(\delta_\ell u) \approx \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}}\sqrt{\frac{2\pi}{\frac{\partial^2 f}{\partial h^2}(h, \alpha, \delta_\ell u))}}\,\exp[-f(h^\star, \alpha, \delta_\ell u)].} \tag{10}$$

i.e

$$\boxed{\mathcal{P}_{\delta_\ell u}(\delta_\ell u) \approx \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}}\sqrt{\frac{2\pi}{\frac{\alpha}{c_2} + 2\alpha^2 e^{2\alpha h^\star}\left(\frac{\delta_\ell u}{\sigma}\right)^2}} \times \exp\left[-\alpha\left(\frac{(h^\star - c_1)^2}{2c_2} - h^\star\right) - \frac{1}{2}e^{2\alpha h^\star}\left(\frac{\delta_\ell u}{\sigma}\right)^2\right]} \tag{12}$$

2.2 Case A: assuming $h^\star = c_1$

As a first approximation, we can consider that :

$$\frac{\partial f}{\partial h}(h^\star, \alpha, \delta_\ell u)\Big|_{h^\star = c_1} = 0 \tag{13}$$

And since

$$\frac{\partial^2 f}{\partial h^2}(h^\star, \alpha, \delta_\ell u)\Big|_{h^\star = c_1} = \frac{\alpha}{c_2} + 2\alpha^2 \left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha c_1} \quad > 0 \tag{14}$$

We can use the Laplace approximation. Eq(12) then becomes :

$$\boxed{\mathcal{P}_{\delta_\ell u}(\delta_\ell u) \approx \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}}\sqrt{\frac{2\pi}{\frac{\alpha}{c_2} + 2\alpha^2\left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha c_1}}} \times \exp\left[\alpha c_1 - \frac{1}{2}\left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha c_1}\right]} \tag{15}$$

2.3 Case B: solving $\frac{\partial f}{\partial h}(h, \alpha, \delta_\ell u) = 0$ via the Lambert W function

From (8),

$$\frac{\partial f}{\partial h}(h, \alpha, \delta_\ell u) = \alpha\left(\frac{h - c_1}{c_2} - 1\right) + \alpha\left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha h} \tag{16}$$

Set to zero:

$$h - (c_1 + c_2) = -c_2 \left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha h} \tag{17}$$

Multiply by $-2\alpha e^{-2\alpha h}$:

$$(-2\alpha)(h - c_1 - c_2)e^{-2\alpha h} = 2\alpha c_2 \left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)} \tag{18}$$

Let

$$u = -2\alpha(h - c_1 - c_2) \quad \Longrightarrow \quad ue^u = 2\alpha c_2 \left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)} \tag{19}$$

Hence

$$\boxed{h^\star = (c_1 + c_2) - \frac{1}{2\alpha} W\left(2\alpha c_2 \left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)}\right)} \tag{20}$$

where $W(x)$ is the Lambert W function (the inverse of $w \mapsto we^w$).

We can now plug the obtained $h^\star$ into Eq(12).

3 Normalized PDF

The p-th order moment of the velocity increment is expressed as

$$\langle |\delta_\ell u|^p \rangle \sim \ell^{\zeta_p}, \quad \text{with} \quad \zeta_p = \min_h \big[ph + 1 - \mathcal{D}^E(h)\big]. \tag{21}$$

Using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h - c_1)^2}{2c_2}, \tag{14}$$

the minimization of Eq. (21) gives

$$\frac{\mathrm{d}}{\mathrm{d}h}\big[ph + 1 - \mathcal{D}^E(h)\big] = 0 \quad \Longrightarrow \quad p = \frac{h - c_1}{c_2}. \tag{22}$$

Hence, the minimizing exponent is

$$h_p = c_1 + pc_2, \tag{23}$$

and inserting this value into Eq. (21) gives

$$\zeta_p = pc_1 + c_2 p^2. \tag{24}$$

In particular, for the second-order moment ($p = 2$):

$$\boxed{\langle (\delta_\ell u)^2 \rangle \sim \left(\frac{\ell}{L}\right)^{2c_1+4c_2}}. \tag{25}$$

We now define a normalized increment

$$\boxed{\delta_\ell \nu = \frac{\delta_\ell u}{\sqrt{\langle (\delta_\ell u)^2 \rangle}} = \delta_\ell u \left(\frac{L}{\ell}\right)^{c_1+2c_2}}. \tag{26}$$

This normalization ensures that

$$\langle (\delta_\ell \nu)^2 \rangle = 1, \tag{27}$$

for all scales ℓ/L, so that all normalized PDFs can be directly compared across scales.

Replacing $\delta_\ell u = \delta_\ell \nu \, (\ell/L)^{c_1+2c_2}$ in Eq. (12) gives the normalized form of the PDF :

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(\delta_\ell \nu) \approx \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}}\sqrt{\frac{2\pi}{\frac{\alpha}{c_2}+2\alpha^2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha h^\star}}} \times \exp\left[-\alpha\left(\frac{(h^\star-c_1)^2}{2c_2}-h^\star\right)-\frac{1}{2}\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha h^\star}\right]} \tag{28}$$

3.1 Case A: assuming $h^\star = c_1$

From (8),

$$f(h^\star,\alpha,\delta_\ell\nu)\Big|_{h^\star=c_1} = \alpha\left(\frac{(c_1-c_1)^2}{2c_2}-c_1\right)+\frac{1}{2}\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha c_1} = -\alpha c_1+\frac{1}{2}\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha c_1}. \tag{29}$$

Moreover,

$$\frac{\partial^2 f}{\partial h^2}(h^\star,\alpha,\delta_\ell\nu)\Big|_{h^\star=c_1} = \frac{\alpha}{c_2}+2\alpha^2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha c_1} \quad > 0, \tag{30}$$

so we can apply the Laplace approximation. Equation (20) therefore becomes

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(\delta_\ell \nu) \approx \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}}\sqrt{\frac{2\pi}{\frac{\alpha}{c_2}+2\alpha^2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha c_1}}} \times \exp\left[\alpha c_1-\frac{1}{2}\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha c_1}\right]} \tag{31}$$

3.2 Case B: solving $\frac{\partial f}{\partial h}(h,\alpha,\delta_\ell\nu)=0$ via the Lambert W function

From (8),

$$\frac{\partial f}{\partial h}(h,\alpha,\delta_\ell\nu) = \alpha\left(\frac{h-c_1}{c_2}-1\right)+\alpha\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha h}. \tag{24}$$

Setting this derivative to zero gives

$$h-(c_1+c_2) = -c_2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha h}. \tag{32}$$

Multiplying by $-2\alpha e^{-2\alpha h}$:

$$(-2\alpha)(h-c_1-c_2)e^{-2\alpha h} = 2\alpha c_2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)}. \tag{33}$$

Let

$$u = -2\alpha(h-c_1-c_2) \quad \Longrightarrow \quad ue^u = 2\alpha c_2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)}. \tag{34}$$

Hence the solution is

$$\boxed{h^\star = (c_1+c_2)-\frac{1}{2\alpha}W\left(2\alpha c_2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)}\right)} \tag{35}$$

Finally, inserting this value of $h^\star$ into Eq.(28) yields the full Laplace-approximated normalized PDF.

4 Entropy of the normalized PDF

We now compute the Shannon entropy of the normalized increment PDF $\mathcal{P}_{\delta_\ell \nu}$ obtained in Eq. (28). By definition,

$$S(\delta_\ell \nu) = -\int_{-\infty}^{+\infty} \mathcal{P}_{\delta_\ell \nu}(x)\, \ln\big[\mathcal{P}_{\delta_\ell \nu}(x)\big]\, \mathrm{d}x \tag{36}$$

References

[1] L. Chevillard, B. Castaing, A. Arneodo, E. Lévêque, J.-F. Pinton, and S. G. Roux, *A phenomenological theory of Eulerian and Lagrangian velocity fluctuations in turbulent flows*, C. R. Physique, **13**, 899–928 (2012).